WJEC Biology Unit 5
Condense Guide
IGC HK Exam

Index

1.1 ⭐ Food Test ⭐

Hazzard	Risk (Action + Body Part)	Control measure
Biuret is an irritant	Could splash onto hands or into eyes when transferring biuret to test tube	Wear gloves/ eye protection
Ethanol is flammable	Could catch fire if used near a Bunsen burner	Ensure all Bunsen burners are turned off before ethanol is used

Reducing Sugar – Benedict's Test

1. Add equal volume of test solution and Benedict's reagent
2. Heat in a warm water bath
3. Positive Result: Blue → Orange / Red precipitate

Non Reducing Sugar

1. Add equal volume of test solution and Benedict's reagent
2. Heat in a warm water bath
3. Observe colour change. If not a reducing sugar → remains blue
4. Add 2 drops of HCl and heat
5. Add 2 drops of NaOH
6. Add Benedicts' reagent
7. Heat in warm water bath
8. Non reducing sugar: Blue → Orange / Red precipitate

Proteins – Biuret Test

1. Add equal volume of test solution and Biuret reagent
2. Cover the top and invert it once. Positive: Colour change from pale blue to purple

Starch – Iodine

1. Add 2cm^3 of test solution and 2 drops of Idodine
2. If starch is present the solution will change colour from yellow brown to blue/ black

Fats and Oils

1. Add equal volume of test solution and alcohol in a boiling tube
2. Shake the tube
3. Pour the mixture into another boiling tube half full of cold water
4. If lipids are present a cloudy white emulsion will form

1.1 ⭐ Determination of glucose concentration using a calibration curve

- Benedict's reagent is used to determine a presence of reducing sugar
- Colour is subjective so using a colorimeter can provide quantitative results

Method:

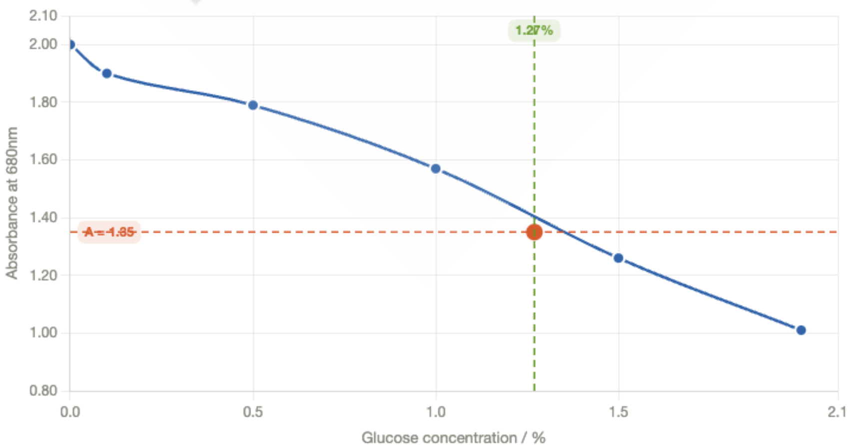
1. Dilute the solution 0f 2% into 0.1%, 0.5%, 1.0%, 1.5% and 2.0% into 5 boiling tubes
2. Mark the unknown solution boiling tube 'U'
3. Add equal amount of Benedicts' reagent to each tube
4. Gently alginate the tubes and place the boiling tubes in water bath for 9 minutes
5. Filter the content for each of the tubes into a clean cuvette
6. Place an addition 5cm^3 Benedict's reagent to a clean cuvette

Create a calibration curve and test the unknown

7. Turn on the colorimeter, set to 680nm and record absorbance
8. Remove the cuvette from the colorimeter and measure absorbance for all 5 concentrations and the unknown solution

Sample Results

Glucose Concentration / %	Mass of glucose in 0.5cm^3 of solution /mg	Absorbance at 680nm
0.00 (Pure Benedict's Reagent)	0.0	2.00
0.1	0.5	1.90
0.5	2.5	1.79
1.0	5.0	1.57
1.5	7.5	1.26
2.0	10.0	1.01
Unknown		1.35



- Use the calibration curve to interpolate the unknown concentration

1.3 ⭐ Determination of water potential by measuring change in mass or length ⭐

- Different water potentials are separated by a selectively permeable membrane
- Water move into solution with lower water potential

Hazzard	Risk	Control measure
Scalpel blades are sharp	May cut skin when cutting cylinders	Cut away from body onto a white tile
Cork borers are sharp	May cut skin when cutting cylinders	The cylinders of tissue must be cut on the chopping board with the force directed downwards

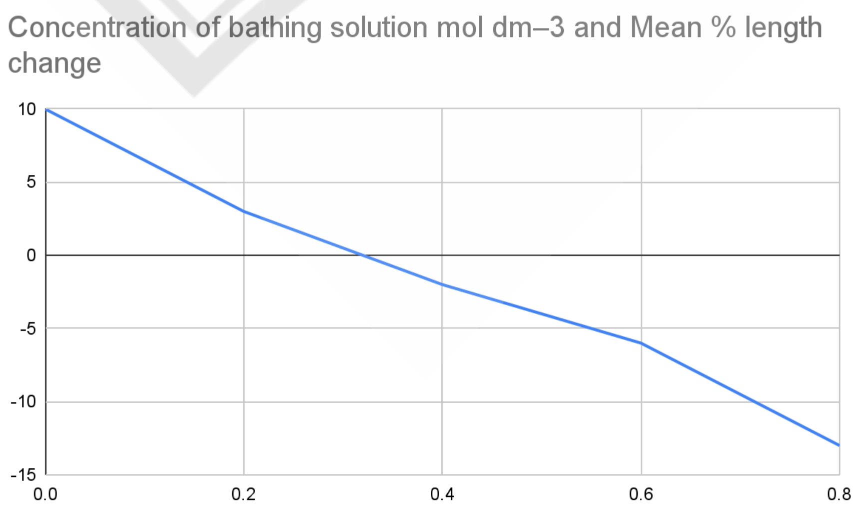
Method

- Remove skin from potato as it makes it waterproof and present osmosis
- Weight the potato
- Place the potato in solution of different concentration and leave it for 20 minutes
- Blot the cylinders to remove water from surface of potato
- Reweight the potato and calculate **percentage change**
- Use **percentage change** as the potato have different starting mass + allow comparison

Sample Results

Concentration of bathing solution mol dm^{-3}	Initial Length / mm	Final Length / mm	Length change / mm	% length change	Mean % length change
0.0	52	57	5	10	10
	49	55	6	12	
	50	54	4	8	
0.2	48	49	1	2	3
	52	2	4	8	
	50	51	1	2	
0.4	58	55	-3	-5	-2
	50	48	-2	-4	
	53	54	1	2	
0.6	50	48	-2	-4	-6
	52	48	-4	-8	
	50	47	-3	-6	
0.8	49	41	-8	-16	-13
	50	43	7	-14	
	52	48	-4	-8	

Plot a graph of Concentration of bathing solution against Mean percentage length change



- No change in length (0%) $\Psi_p = 0$ $\therefore \Psi_{cell} = \Psi_s$ at this graph is 0.3

Molarity / mol dm^{-3}	Solute potential /kPa
0.05	-130
0.10	-260
0.15	-410
0.20	-540
0.25	-680
0.30	-860
0.35	-970
0.40	-1120
0.45	-1280
0.50	-1450

- If time to leave the potato in solution is too short results are still valid but error are larger so data are less accurate
- Dot to dot (not line of best fit)

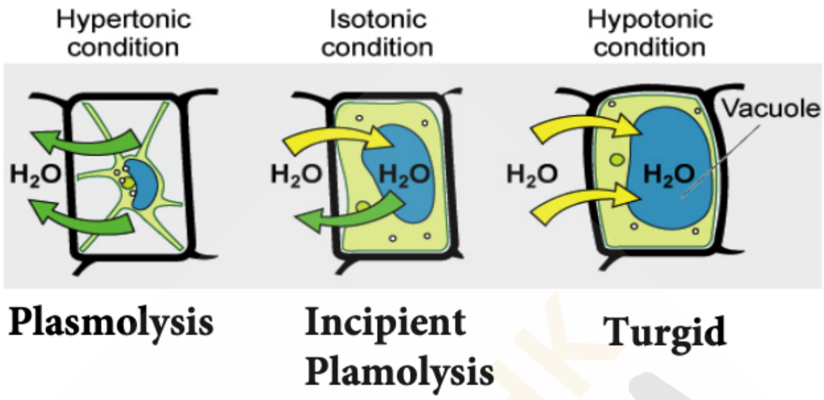
Potential WJEC 2026 Experiment

- Measure the change in mass / time / volume of solution / length / pH of solution
- Changing the sweetness of potato affects the intercept at x axis. A hypothesis states that horizontal axis would be at a higher concentration for sweet potato as sweet potato have a higher concentration of dissolving suagrs

Other Notes

- Definition: Water potential is the tendency of water molecules to move into / out of a cell or solution.
- Solute potential ψ_s: Measures how easy water molecules can move out of a solution
- Pressure Potential ψ_p: Measures the pressure exerted on the cell contents by the cell wall
- Raw potato, ✓ Osmosis – Differently permeable cell membrane present
- Peeled Potato, ✓✓ – High rate of Osmosis, because larger Surface Area for Osmosis
- Boiled Potato, X Osmosis – Damage cell membrane -> Fully Permeable
- Unpeeled Potato, X – Impermeable in H_2O

1.3 ⭐ Determination of solute potential by measuring the degree of incipient plasmolysis ⭐



At incipient plasmolysis:

- No <u>net</u> movement of water
- <u>Cell membrane is withdrawn from cell wall</u>
- Cell content <u>exert no pressure</u> on cell wall

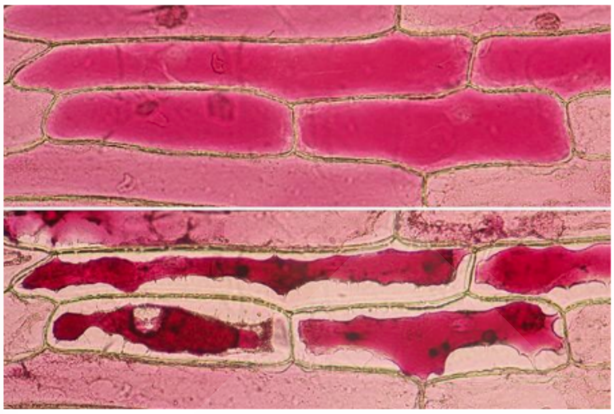
$$\Psi = \Psi_s + \Psi_p$$

Ψ_p = 0 at incipient plasmolysis
So $\Psi_{clll} = \Psi_s$

Hazzard	Risk	Control measure
Scalpel blades are sharp	May cut skin when cutting cylinders	Cut away from body onto a white tile

Method

- Different concentration of solidum chloride solution (0.2, 0.4, 0.6, 0.8 mol dm^{-3}) are set up
- Extract the upper epidermis of onion leaf from mesophyll
- Spread the tissue out on a microscope so that it won't fold
- Add two drops of solution and apply to cover slip
- Use x10 and x40 objective lens to count the number of cells that are turgid and plasmolysed
- Record data in a table and plot a graph for % cell plasmolysed against concentration of solution
- Interpolate the concentration of solution where % plasmolysed cell is 50%
- Use the same table as above to determine the solute potential



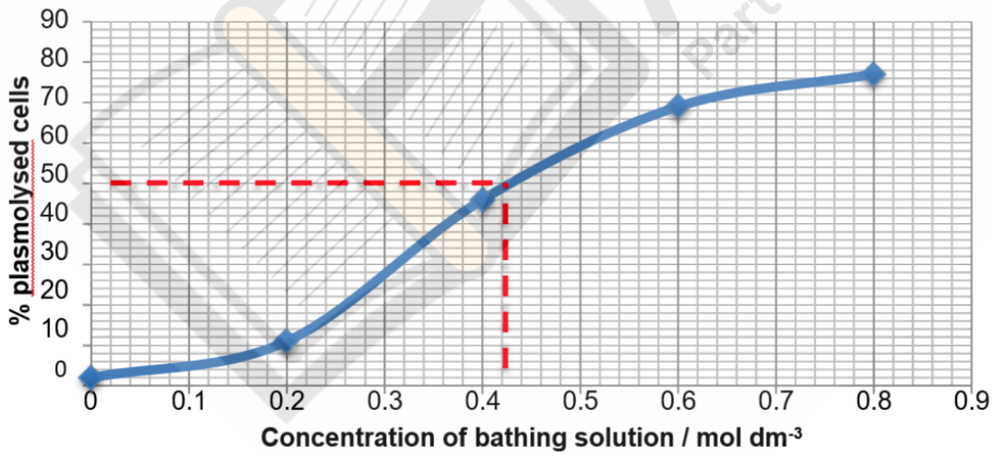
Turgid Cells

Plasmolyed Cells

- To ensure the same cell is not counted more than once, student can move the slide in the same direction across the field and use this as a tracking system
- If the cell is on the edge, only count it if it is at least half visible

Sample Results

Concentration of bathing solution / mol dm^{-3}	Number of cells in field of view						Total		
	1		2		3				
	plasmolysed	turgid	plasmolysed	turgid	plasmolysed	turgid	plasmolysed	turgid	% cells plasmolysed
0	1	30	0	32	1	36	2	98	2
0.2	4	28	2	32	5	29	11	89	11
0.4	20	15	10	19	16	12	46	54	46
0.6	20	9	25	11	24	11	69	31	69
0.8	19	7	34	8	24	9	77	23	77



Molarity / mol dm^{-3}	Solute potential /kPa
0.40	-1120
0.45	-1280

1.3 Investigation into the permeability of cell membranes using beetroot

- Motion of molecules within the membrane is affected by temperature
- Heating the membrane can cause membrane to become more permeable
- Heat can denature the membrane

- Beetroot contain betalain, a bright red, water soluble pigment in the cell vacuoles
- If the cell membranes are damaged, the pigment can escape from the cell and can be detected in an aqueous medium
- Beetroot must be raw, not cooked so the cell membrane are not damaged

Hazzard	Risk	Control measure
Scalpel blades are sharp	May cut skin when cutting cylinders	Cut away from body onto a white tile

Method

- Cur 5 piece of beetroot and wash the pigment released during cutting
- Place 5cm^3 of distilled water in a test tube and place in water bath to equilibrate for 5 minutes
- Place the beetroot in each test tube for 30 minutes
- After 30 minutes, shake the test tubes gently to make sure any pigment is well mixed in the water and remove the beetroot cores
- Place it in a colorimeter and measure absorbance
- Repeat the steps for different water bath temperature and plot a graph of absorbance against temperature

Sample Result of a colorimeter – measuring transmission of light at 530nm

Temperature (°C)	Observation	Colorimeter reading (% transmission of light)			
		Repeat 1	Repeat 2	Repeat 3	Mean
0	Clear	100	98.5	99.0	99.2
22	Very pale pink	93.9	95.0	96.0	95.0
42	Very pale pink	80.1	77.0	76.9	78.0
63	Pink	26.3	29.9	31.0	29.1
87	Dark pink	0.7	0.7	1.0	0.8
93	Red	0.0	0.1	0.0	0.0

WJEC 2026 Possible Variation

- Investigating the effect of alcohol or detergents on membrane permeability
- Identifying systematic and random variables (errors)
  Systematic: Stop clock that is running slow, so there will be an increase in the amount of pigment that leaks
  Random: Different piece of equipment (balance) measures data slightly differently
- Plotting error bars to assess the variation in repeats

3.2 Investigation into the separation of chloroplast pigments by chromatography

- Photosynthetic (chloroplast) pigmenters are located on membranes of thylakoids and grana
- Photosynthetic pigments are used for harvesting light in the light-dependent reaction for photosynthesis and transferring its energy to the light independent ration
- Chloroplast Pigments

  Chlorphylls
  - <u>Chlorophyll a</u> is the most commonly found pigment in all photosynthetic organisms
  - <u>Chlorophyll b</u> is found in flowering plants
  - *Phaeophytin: a breakdown product of chlorophyll a molecule, lacking the central magnesium ion is seen in flowering plants and in purple sulphur bacteria*

  Carotenoids
  - <u>Carotenes</u>: α- and β-carotene are orange but lycopene is bright red is found in tomatoes
  - <u>Xanthophylls</u>: such as lutein and zeaxanthin appear yellow

Hazzard		Risk	Control measure
Propaponone Petroleum either	May cause eye dammage	Macerating leaf material; Pouring solvent for chromatography	Eye protection
	May degrease the skin	Macerating leaf material; Pouring solvent for chromatography	Wear gloves
	Inhalation may exacerbate respiratory problems, including asthma	Macerating leaf material; Pouring solvent for chromatography	Work in fume cupboard
	Fire hazards	Accidental ignition	Work in fume cupboard

Method

Preparing the pigment solution

1. Chop 2g of leaf material, add a pinch of sand and 5 cm^3 of propanone in a mortar and grin the leaf fragment to a slurry. And transfer it into a boiling tube
2. Add 3cm^3 of instilled water, shake vigorously and stand for 8 minutes
3. Add 3cm^3 of petroleum ether mix by gentle shaking and allow layers to separate
4. Collect the upper petroleum ether layer which contains chloroplast pigment and using a pipette, transfer to a vial

Preparing the chromatography paper

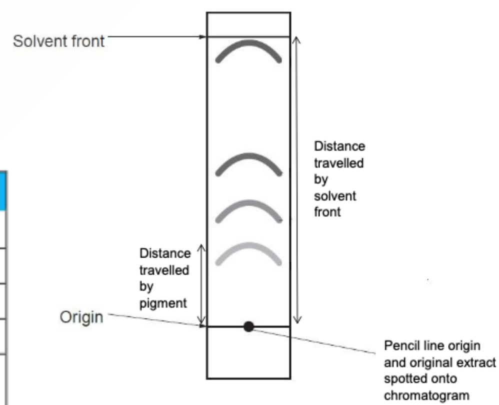
1. Draw a pencil line across the chromatography paper approximately 2cm from one end
2. Draw chloroplast pigment solution in a capillary tube and put a small spot in the center of the pencil line. Ensure that the capillary tube does not pierce or tear the chromatography paper
3. Dry the spot as quickly as possible to prevent it spread
4. Repeat steps 2 and 3 until there is a small but more intense spot of pigment

Running the chromatogram

1. Place freshly-made 1:2 propanone:petroleum ether solvent mixture in a boiling tube until it is approximately 5mm deep
2. Slide the chromatography paper into the boiling tube so that one end is below the surface of phenol but the spot its above so not touching it
3. Hold the chromatography paper in place with the stopper, folding the paper over the rim of boiling tube at the top
4. Leave the boiling tube until the solvent has climbed up the paper to within 10mm of the top. Mark a pencil line across the paper indicating the solvent font.
5. Mark the position of the top of each pigment spot with a pencil

Identifying the pigments

$$R_f = \frac{\text{distance travelled by pigment}}{\text{distance travelled by solvent front}}$$



Published data for 1 propanone : 2 petroleum ether

Spot colour	Pigment	R_f
yellow	ß-carotene	0.96
grey	phaeophytin	0.70
blue-green	chlorophyll *a*	0.58
green	chlorophyll *b*	0.48
yellow-brown	xanthophyll	0.44 (TLC) 0.75 (paper chromatography)

Possible 2026 WJEC Experiment

- Use different solvent // Compare pigment in young and senescent deciduous leaves // Compare pigments in sun and shade leaves

3.2 Investigation into the effect of light on the rate of photosynthesis

$$\text{carbon dioxide + water} \xrightarrow{\text{Light}} \text{glucose + oxygen}$$

- Measure of a pH colour change due to carbon dioxide acidic properties
- When CO_2 is being absorbed in photosynthesis, it becomes less acidic
- When CO_2 is dissolved in water, carbonic acid is produced, and it dissociates releasing hydrogen ions, lowering the pH of water

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

- Photosynthesis removed CO_2 from solution, H^+ concentration decreased so pH increases

Hydrogen Carbonate Indicator Colour Change – Photosynthetic Protoctistan

$$\underset{\text{Yellow}}{\text{Acid}} \rightleftharpoons \underset{\text{Red}}{\text{Neutral}} \rightleftharpoons \underset{\text{Purple}}{\text{Alkali}}$$

Purple: Rate of Photosynthesis > Rate of Respiration
Yellow: Rate of Photosynthesis > Rate of Respiration
Red: Rate of Photosynthesis = Rate of Respiration (Compensation Point)

Hazzard	Risk	Control measure
Solid calcium chloride is an irritant to skin and eyes and in inhaled	Making calcium chloride solution	Solid to be weighted in fume cupboard; Students to be given solution.
Gas accumulation in culture vessel could cause the glass to break	During period of algal culture	Ensure cotton wool stopper allows ventilation; Stand culture vessel in deep tray
Excess heat from lamp may cause burns	When decanting from culture vessel	Ensure no contact with skin

Method

Making algal balls

1. Stir a mixture of $5cm^3$ Scenedesmus culture and $3cm^3$ 3% sodium alginate solution gently
2. Draw the mixture into a $10cm^3$ syringe barrel
3. With constant pressure on the plunger, drop the mixture, one drop at a time, into $200cm^3$ calcium chloride solution
4. Leave the balls for 20 minutes
5. Strain the balls through the tea strainer
6. Return the balls to the beaker and swirl them in distilled water
7. Repeat step 5 and 6 twice more
8. Use immediately or store at 4°C, but bring to room temperature for approximately 20 minutes before use

Running the experiment

1. Place 20 algal balls in a viral
2. Add $10cm^3$ hydrogen carbonate indicator
3. Place the vials at a distance from a light source
4. After a given time assess the pH of the indicator in the vial using teh colour chart or read its absorbance at 550m in a colorimeter

Sample Results

Distance of vial from lamp / cm	Colour of indicator after 1 hour	Absorbance after 1 hour / a.u.	pH after 1 hour
10	purple	0.92	9.0
30	magenta	0.81	8.8
50	red	0.56	8.4
70	orange	0.39	8.2
90	yellow	0.25	8.0

Possible WJEC 2026 Practical

- Varying light intensity to determine colour change
- Cover the vials in different colours filtered to expose algae at different wavelengths. In order that the algae receive the same light intensity readings must be taken with a light meter with the filters over the probe, to find distance for each wavelength which has the same light intensity

3.2 Investigation into the role of nitrogen and magnesium in plant growth

- Plants require nitrogen in the form of nitrate in absorbed by the roots to make amino acids, chlorophyll and nucleotides
- Magnesium ions are also absorbed by the roots of a plant and use as a component of chlorophyll

Hazzard	Risk	Control measure
Sach's culture solution can be an irritant	Very low risk of irritation to the eye or skin.	Wear goggles. Wash skin thoroughly if solution comes into contact with skin.

Method

1. Select equal size barley seedlings.
2. Set up 5 test tubes with Sach's complete culture solution, 5 test tubes with Sach's culture solution lacking nitrate and a further 5 test tubes with Sach's culture solution lacking magnesium.
3. Solutions should be topped up when necessary and completely replaced weekly.
4. All 15 barley seedlings should be placed in the same conditions for example light and temperature.
5. After a month examine the seedlings, record any differences between them and measure the length of the roots and shoot.
6. Dry the seedlings in an oven and record the dry mass

Sample Results

Lacking in nitrogen – poor growth, plants short and spindly and chlorosis (yellowing), especially in the older leaves. The young leaves at the tip may be green but small.

Lacking in magnesium – interveinal chlorosis.

Possible Questions

Why use Dry mass → Dry in oven until no weight change to get rid of water

- Dry mass = Organic Mass (carbohydrate, lipid, protein)
- As some organism might live in water

Why top up solutions daily?

- Volume must be kept constant

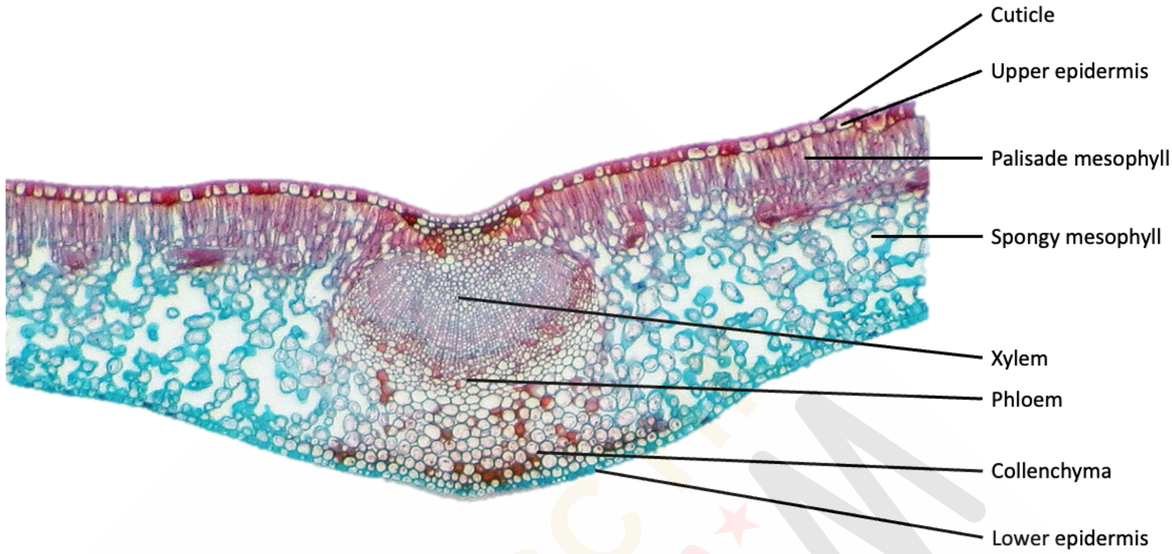
Why change the solution weekly?

- Allow conc of mineral same and ensure the minerals are sufficient for growth

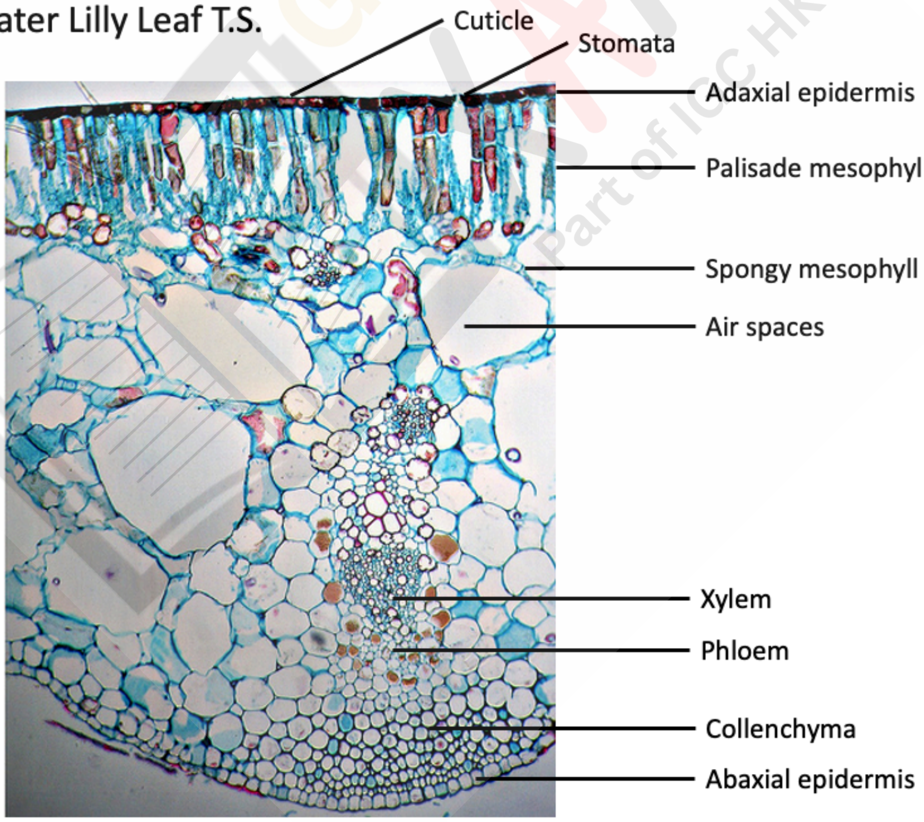
Lack of Nitrogen - No Amino acid / protein → cannot growth / no new cell growth - No Nucleotide - No Chlorophyll → yellow → chlorosis → very little photosynthesis	Lack to Magnesium - No Chlorophyll → yellow → chlorosis → little photosynthesis as some leaf might have chlorophyll	Lack of Calcium - Weakens cell wall

3.2 Cross-section of a leaf

Low power Leaf



Low power Water Lilly Leaf T.S.



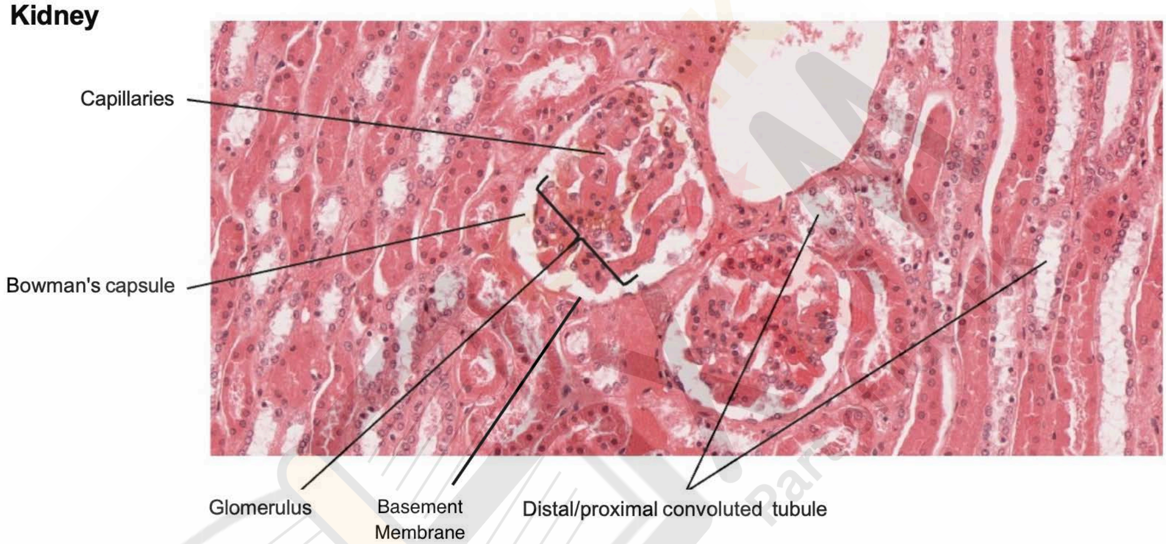
3.8 ⭐ Dissection of a mammalian kidney ⭐

- Kidney functions are to remove nitrogenous wastes urea, from body to balance pH and ion of blood

Hazzard	Risk	Control measure
Dissecting instruments are sharp	Can pierce or cut the skin	Care with use

Required Label: Bowman's capsule; glomerulus; capillaries; distal/proximal convoluted tubule; basement membrane

Kidney



4.4 ⭐ Investigation of a continuous variation in a species ⭐

Hazzard	Risk	Control measure
Ivy leaves may be allergenic and generate contact dermatitis	Skin contact on handling may occur	Avoid skin contact
Berries are toxic	If leaves are collected in autumn, berries may be present	Avoid ingesting berries

- T test shows the difference in two sample means
- Formula to calculate standard deviation from sample

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

- Formula to test that the means are different using Student's t test.

$$t = \frac{\left|\bar{x}^1 - \bar{x}^2\right|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

Where,

$\left|\bar{x}^1 - \bar{x}^2\right|$ = the difference in mean values of sample 2 and sample 2

s_1^2 and s_2^2 are the squares of the standard deviation of the samples

n_1 *and* n_2 are the number of readings in each sample.

Sample Data

Maximum width of ivy leaf (mm) grown in	
shade	sun
17	7
16	11
18	8
21	8
19	9
20	10
17	9
19	10
18	9
17	10
18	10
18	11
16	12
15	18
18	10
mean = 17.8	mean = 10.1

The Student t test

1. Formulate a null hypothesis: there is no statistically **significant** difference between the **mean** maximum widths of the two populations of ground ivy leaves growing in the sun and shade.
2. Process data

Ivy leaves growing in shade			Ivy leaves growing in the sun		
Maximum width / mm	Deviation from mean $(\bar{x} - x)$	Deviation from mean² $(\bar{x} - x)^2$	Maximum width / mm	Deviation from mean $(\bar{x} - x)$	Deviation from mean² $(\bar{x} - x)^2$
17	0.8	0.64	7	3.1	9.61
16	1.8	3.24	11	-0.9	0.81
18	-0.2	0.04	8	2.1	4.41
21	-3.2	10.24	8	2.1	4.41
19	-1.2	1.44	9	1.1	1.21
20	-2.2	4.84	10	0.1	0.01
17	0.8	0.64	9	1.1	1.21
19	-1.2	1.44	10	0.1	0.01
18	-0.2	0.04	9	1.1	1.21
17	0.8	0.64	10	0.1	0.01
18	-0.2	0.04	10	0.1	0.01
18	-0.2	0.04	11	-0.9	0.81
16	1.8	3.24	12	-1.9	3.61
15	2.8	7.84	18	-7.9	62.41
18	-0.2	0.04	10	0.1	0.01
mean = 17.8		sum = 34.4 s_{shade} = 1.57	mean = 10.1		sum = 89.75 s_{sun} = 2.53

$$t = \frac{|\bar{x}_{shade} - \bar{x}_{sun}|}{\sqrt{\frac{s^2_{shade}}{n} + \frac{s^2_{sun}}{n}}}$$

$$t = \frac{|17.8 - 10.1|}{\sqrt{\frac{6.41}{15} + \frac{2.46}{15}}} = \frac{7.7}{0.77} = 10.0$$

3. Degree of Freedom = (df) = (15-1) + (15-1) = 28
4. For df = 28 and level of significance, p = 0.05, the critical value of t = 2.048
5. Comparing the critical value with the calculated value: <u>the calculated value is greater than the critical value so the null hypothesis is rejected (10>>2.048)</u> at the 5% level of significance, or p = 0.05
6. Conclusion: the maximum mean widths of the two populations <u>are significantly different</u> at the 5% level of probability. The maximum width of ivy leaves is greater growing in the shade than growing in the sun.

Risk Assessment

- Hazzard: Irritant (etc) [if cannot think about it, organism might cause allergy to human]
- Risk: Action + Body Part
- Control Measure: Wash hands, Lab coat, Wear gloves

Range Bar

Provide provide information on <u>reliability</u> of the readings

Use percentage Change in mass instead of actual mass

- DIfferent starting mass (1)
- Allows <u>**comparison**</u> (1)

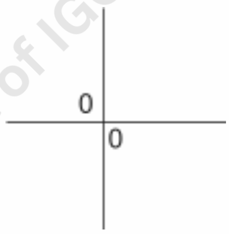
‘Must Get’ mark from Practical Paper

Table:

- Table with heading and **be specific** [eg: mass → ✓ mass of potato] (do not include irrelevant colum)
- Units (if use short term ✓min X mins)
- Same significant figure (2sf) down and across

Graph

- More than ½ space
- Labels (be specific eg: **mean** percentage change)
- Units
- Axis
- Plots correctly
- Line (dot to dot)
- Range bar



T Test

- <u>T Test Null Hypothesis: There is **no statistically significant difference** between the **means** of [height / mass / description] of the two samples and that any difference is due to chance.</u>
- Find t value
- T Test Degree of Freedom: Number of Sample 1 + Number of Sample 2 – Number of Samples (so 2 in this case)
- Probability: 5% (or 0.05)

This means there are only 5% probability that is due to chance

- Find critical value
- If <u>t value > critical value</u> → reject null hypothesis → not due to chance → due to other factors: (for example lack of Mg^{2+})
- If t value < critical value → accept null hypothesis → any difference is due to chance

Maths: higher the t value / chi^2 value, the smaller the probability it will occur (nothing related to probability so do not use the binomial hypothesis testing mindset as it is literal opposite)

- To be more accurate use a smaller percentage
- Use 1 more sf than the sf number in the significance level chart give

Variables

- Independent variables answer the question "What do I change over a suitable range?"
- Dependent variables answer the question "What do I observe?"
- Controlled variables answer the question "What do I keep the same to prevent their influence on the effect of the independent variable on the dependent?"

Improving experimental design

Questions about…	What this means…	How to improve the experiment…
Subjectivity of results	Some of the experiments that you do in biology rely on you to look for colour changes in order to get results e.g. starch/amylase and iodine experiments. Different people may judge colour differently, so your 'end-point' may be different to someone else's – this affects the reproducibility of an experiment.	A colorimeter is a piece of equipment designed specifically to register changes in the absorbance/transmission of light at different wavelengths. Instead of judging a colour, you can read exactly what the absorbance is at a particular wavelength.
Reliability	Reliability is to do with the consistency of your data – how similar your repeat readings are.	To increase the reliability of an experiment, more readings should be taken and a mean calculated. Range bars on a graph can be used show you the reliability of your experimental data If there is clearly an anomalous reading, it is justifiable to remove it before calculating the mean, **as long as you explain what you have done and why**. Do not simply circle the anomaly – this may be OK in another subject but in Biology you need to explain what you are doing and why!

Questions about…	What this means…	How to improve the experiment…
Confidence	How confident are you that your results match what really should be happening?	Repeating an experiment and getting the same/similar results shows that you can have a high level of confidence in the results. Another way of improving your confidence in any conclusions is to change the range of your independent variable. If your experiment shows that an enzyme works better at pH 6 than any other pH you chose as to use, then using smaller range intervals e.g. 5.4, 5.6. 5.8, 6.0, 6.2, 6.4 and so on would allow you to be even more confident in your conclusions.
Accuracy	Accuracy is to do with measurement. You are always limited to the level of precision of your apparatus.	If you are using a syringe with a barrel calibrated in cm^3 then a reading of volume is only accurate to the nearest 0.5 cm^3. Using a syringe calibrated to 0.1 cm^3 increases the accuracy. You may need to change apparatus completely to improve your accuracy, if a measuring cylinder is calibrated to a maximum precision of 1 cm^3 then using a burette calibrated to 0.1 cm^3 is a huge improvement.
Control experiments	**These are different to control variables!** An experimental control is a way of proving that it's the biological material that is bringing about the changes that cause your results. It proves that organisms need to be living and that their enzymes need to be functioning in order to carry out the reaction you're monitoring.	Most experimental controls are pretty straightforward; you should **boil and cool** any living organism or enzyme that you are using and then use it in an experiment. This will kill the organism and denature all of its enzymes. These control experiments should not work – if they do that shows you that it wasn't the enzyme or organisms that was giving you the result in the first place!

Questions about…	**What this means…**	**How to improve the experiment…**
Validity	Validity is concerned with whether you are justified in drawing conclusions from your experimental results.	In a <u>laboratory-based experiment</u>, your decision about how valid your results are will be linked to how well you have regulated your control variables. If you have thought about your control variables properly then you should have a high level of confidence in your results. In an <u>environmental investigation</u>, producing valid results is much more difficult. There are so many factors that cannot be effectively controlled, e.g. the weather, that this can cause problems with the validity of our conclusions. If you are comparing two or more sites in an ecological investigation, then you need to try and make the results valid by ensuring that the sites and the way the investigation was carried out are as similar as possible. The factors that need to be controlled are usually abiotic factors such as: • the time of year the investigation is carried out • the time of day the investigation is carried out • altitude (the height of the test sites above sea level) • aspect / slope (in the northern hemisphere, northern slopes face away from the sun and southern ones are much sunnier) • pH of the soil or water in the environment

Tabulating data

- full column headings, not simply shortened to words like 'concentration' or 'temp'
- appropriate recording of readings, e.g. times recorded to the nearest second, with the same number of decimal places throughout table except for the mean which can be to a maximum of 1 decimal point more than your dependent variable recordings

Independent variable / unit	Dependent variable / unit			
	Trial 1	Trial 2	Trial 3	Mean
Value 1				
Value 2				
Value 3				

Concentration of amylase / mol dm^{-3}	Time taken for the iodine to stay yellow/brown / seconds			
	Trial 1	Trial 2	Trial 3	Mean
0.1	360	350	340	350.0
0.2	280	250	260	263.3
0.5	150	290	140	145.0
1.0	60	50	60	56.7

= this result is anomalous and is not used for the calculation of a mean

Graphing skills

- **Correct axis choice**
- **Correct labelling of axes**
  vertical axis should be a 'Mean….' recording
- **Correct use of scale**
  You also need to make sure that both axes have a number marked on each axis in the corner of the graph – this does not need to be '0'.
- **Correct placing of data points**
- **Use of a ruler to join data points**
  In biology you will always be expected to join data points with a straight, ruled line. You should only use a line of best fit if you are specifically told to do so
- **Only join points for which you have data, do not force through origin**
- **Use of range bars**
  Range bars give a good indication as to the reliability of your data.

Analytical skills

- **Identify any trend in the results**

  By looking at the graph you have drawn you should be able to describe the shape of the graph drawn. Is it a positive correlation? Does the graph increase and then level off? Does the graph have a bell-shape? Remember to quote figures from your data that reflect the graphs gradient and any points where the graph levels off/falls.

- **Comment on the accuracy of the experiment**

  Accuracy is all to do with whether your results are close to the true value you would expect. Carrying out repeats of the experiment would allow you to see the spread of your data, the more repeats you have that are consistent with each other, the more confidence you can have that your results are accurate.

- **Comment on the consistency of your data**

  Use your range bars to comment on this. If your range bars are small then you have consistent data, if they are wide then you have inconsistent/unreliable data. Errors bars from adjacent points on straight line graphs shouldn't ideally overlap if your data is reliable and you should comment on this where it occurs. They will probably overlap if the curve is bell shaped (they have to if you think about it!) so this overlap doesn't really warrant a comment.

- **Comment on the precision of the readings**

  While you are carrying out your experiment you should always be thinking about how precise your data collection is. Precision relates to measurement of any of your variables but especially your dependent one. Is the way you are measuring results prone to error? Can you think of a more precise method for collecting data? Maybe you can use more precise equipment? A burette is far more accurate than a measuring cylinder, for example.

- **Comment on any subjectivity in your results.**

  You will usually be able to measure a dependent variable using a relatively precise method, such as with a timer or a piece of equipment such as a measuring cylinder. Sometimes, however, you will be relying on something like a colour change to collect your dependent result; this is an acceptable method for data collection, but it has an issue, it is far to subjective to be truly useful. People may judge the same colour in different ways so we need a more precise measurement; a colorimeter is the most common, precise way of judging shades or depth of colour but you wouldn't use it if you were using universal indicator to judge a change of pH, here an electronic pH probe will give you a digital readout that is more useful than the results you'd get from a colorimeter.

- **Suggest improvements for any inaccuracies identified**

  Having spotted areas where the experiment can be improved it should be a pretty natural thing to go on to say how you can make the experimental design better – this is where you need to think about controlled variables - maybe these haven't been as well controlled as they might've been. If you have been using a beaker of warm water to keep things warm and the temperature has fallen slowly over time, then using an electronically controlled thermostatic water-bath to better regulate such an important dependent or control variable. If you have been judging the result of an experiment by using something as subjective (vague) as a colour change, then an obvious place where improvements can be made is by using a colorimeter.

  You may also be asked about a **control experiment** – this is not the same as a controlled variable! A suitable control for an enzyme experiment is to use a **boiled <u>and cooled</u>** enzyme material in place of an active one. If you have been using an enzyme sourced from a living thing, like yeast or potato, then <u>boil and cool the tissue/material</u> – that should denature any enzymes present inside its cells before you extract them and prove that it's the enzyme in its active form that is causing the reactions that you are investigating.

- **Give an explanation of results using relevant and sound biological knowledge**

  This is where you bring in all of your knowledge from theory lessons to explain the results of your experiment. The more detail it contains the better, but you may potentially get penalized for 'throwing it all in' and including irrelevant knowledge.

- **Draw a suitable conclusion**

  This is where you sum up all of your results into one brief statement. It's at this point that you may also need to say whether any hypothesis or prediction that was made was correct or not.

Risk Assessments

- identify the hazard (what it is that is dangerous)
- identify the risk
- state what you would do to minimise the risk.

Difference between a hazard and a risk:

- The hazard is the item that might cause you harm, e.g. a scalpel blade is sharp, or ethanol is flammable.
- The risk describes the activity that might cause the harm, they should always contain an action and a body part e.g. when using a scalpel in dissection the blade might slip and you could cut your skin, or ,if you are performing a starch test and a flame has not been extinguished before pouring it, then ethanol could catch fire and cause burns to your hands or skin.

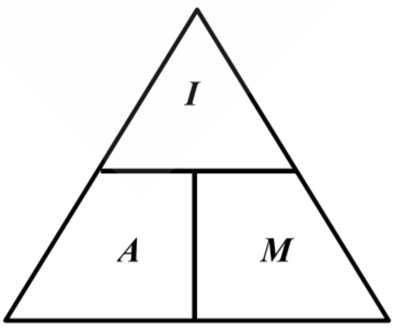
Hazard	Risk	Control Measure
What is the danger?	*How is it dangerous **and** when?* This **must** include an action and a reference to a body part or injury.	*How do you minimise the risk / stay safe?*

Examples

Hazard	Risk	Control Measure
Methylene blue is an irritant	Methylene blue could spill onto your skin when you are transferring it from the flask to the tube	Wear gloves and wash hands in case of contact with methylene blue
Scalpels are sharp	You could cut into your fingers when you are slicing pieces of potato to their correct length	Always cut down onto a white ceramic tile. Cut away from yourself.

Microscopy

Details in igchk-bio-u5-wjec



I = Size of image
A = Actual size of object
M = magnification.

IGC HK Exam - WJEC

GCE Biology Unit 5 Condense Notes

Though the resources are comprehensive, they may not cover every aspect of the specification and do not represent the depth of knowledge required for each unit of work.

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